

$$\begin{aligned}
& \wedge op\_seq' = \\
& \quad [op\_seq \text{ EXCEPT } \\
& \quad \quad ![t] = \\
& \quad \quad \quad Append(@, [type \mapsto \text{"W"}, key \mapsto op.key, val \mapsto op.val])] \\
& \wedge pc' = [pc \text{ EXCEPT } ![t] = pc[t] + 1] \\
& \wedge \text{UNCHANGED } \langle client\_dep, \\
& \quad \quad client\_last\_commit, \\
& \quad \quad router\_ct, \\
& \quad \quad shard\_ct, \\
& \quad \quad tx\_status, \\
& \quad \quad tx\_snapshot, \\
& \quad \quad tx\_commit\_time, \\
& \quad \quad store \\
& \quad \rangle \\
CommitTx(t) & \triangleq \\
& \wedge tx\_status[t] = \text{"Active"} \\
& \wedge pc[t] > Len(Workload[t].ops) \\
& \wedge \text{LET } IsConcurrent(t\_other) \triangleq \\
& \quad tx\_status[t\_other] = \text{"Committed"} \wedge \\
& \quad tx\_commit\_time[t\_other] > tx\_snapshot[t] \\
& HasConflict \triangleq \\
& \quad \exists t\_other \in AllTxns : \\
& \quad \quad \wedge IsConcurrent(t\_other) \\
& \quad \quad \wedge \exists k \in tx\_checkset[t] : \\
& \quad \quad \quad \exists v \in store[k] : \\
& \quad \quad \quad v.writer = t\_other \wedge v.cts = tx\_commit\_time[t\_other] \\
\text{IN IF } HasConflict & \\
& \quad \text{THEN } \wedge tx\_status' = [tx\_status \text{ EXCEPT } ![t] = \text{"Aborted"}] \\
& \quad \wedge \text{UNCHANGED } \langle client\_dep, \\
& \quad \quad client\_last\_commit, \\
& \quad \quad router\_ct, \\
& \quad \quad shard\_ct, \\
& \quad \quad tx\_snapshot, \\
& \quad \quad tx\_checkset, \\
& \quad \quad tx\_buffer, \\
& \quad \quad tx\_commit\_time, \\
& \quad \quad pc, \\
& \quad \quad store, \\
& \quad \quad op\_seq \\
& \quad \rangle \\
& \quad \text{ELSE } \wedge shard\_ct' = shard\_ct + 1 \\
& \quad \wedge tx\_commit\_time' = [tx\_commit\_time \text{ EXCEPT } ![t] = shard\_ct'] \\
& \quad \wedge tx\_status' = [tx\_status \text{ EXCEPT } ![t] = \text{"Committed"}] \\
& \quad \wedge store' =
\end{aligned}$$